Descriptor Similarity Does Not Predict Array Response: Chemical Warfare Agent Simulants Tested Against Seventeen Authentic Agents

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Abstract

Chemical warfare agent detection is developed and qualified on simulants, and we asked what measurement establishes that a simulant reproduces the agent it stands in for. A 29-dye colorimetric array was used to measure seventeen authentic agents, spanning nerve, blood, choking, vesicant and A-series compounds, at five concentrations in two optical channels, and the resulting response patterns were compared with the molecular-descriptor distances on which simulant selection in practice rests. Agreement was tested by exact Mantel permutation within each run, by exact enumeration of all 680 possible simulant designations in descriptor space, and against permutation nulls built for each comparison. The array separates compounds in all six run-channel cells, with p from 4.6×10^{-8} to 0.025, so the measurement is sound. Descriptor distance nonetheless fails to predict response distance: twelve exact Mantel tests and six

combinations of them leave nothing below $q = 0.05$. Dimethyl methylphosphonate, the standard G-agent simulant and the benchmark compound for decontamination efficiency, is anti-correlated with soman and cyclosarin in the ultraviolet channel, and its correlation with sarin falls from $+0.77$ at 500 ppm to $+0.36$ at 10 ppm, the end at which point detectors operate. Across the seventeen agents measured, exactly one designated simulant has ever been measured beside them.

Introduction

Nearly every chemical warfare agent detector in the literature is developed, optimised and qualified on compounds that are not chemical warfare agents.[1,2] Dimethyl methylphosphonate stands in for the G-series nerve agents,[3] dipropylene glycol monomethyl ether for training detectors,[4] and methyl salicylate for vapour penetration through protective ensembles.[5,6] The substitution is unavoidable, because the agents are lethal at concentrations where handling them requires a licensed facility, and it is load-bearing: a sensor reported as selective for sarin has, in most cases, never been exposed to sarin.

What licenses the substitution is a claim of similarity. The compound chosen is said to be agent-like, and similarity is read from molecular descriptors: vapour pressure, dipole moment, frontier orbital energies, substructure and shape. The claim that matters for a detector is different. It is that the substitute produces the response the agent would produce in the same device. Those two claims are connected only if descriptor similarity transfers to response similarity, and that transfer has not been measured.

It has not been measured because measuring it requires the agents. Response data on authentic agents exist in few places, and the compounds used as simulants are rarely run alongside them. A recent survey of the field found no standard for chemical warfare agent simulants at all, and recorded that they have generally been selected on the basis of their historical use.[7] Our institute is an OPCW designated laboratory, which identifies and assays authentic agents under an accredited procedure, so the comparison can be made on the

agents themselves.

This work reports that measurement. A 29-dye colorimetric array was used to record the responses of seventeen authentic agents at five concentrations in two optical channels, and those responses were compared against the molecular-descriptor distances that simulant selection relies on. The array separates compounds in every run and channel, so the measurement is sound. Descriptor distance nonetheless carries no detectable information about response distance. Dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, is anti-correlated with two of the four G agents, and its resemblance to sarin weakens as the concentration falls towards the range where point detectors operate. Where the designation record and a regulatory toxicity model might have supplied the missing evidence, both are empty for the same compounds.

This is not a new sensing method, and we propose no replacement compound. It is a measurement of whether the substitutes the field already relies on carry the property they are relied upon for.

Results and Discussion

The array separates the agents

A comparison between descriptor space and response space is worth making only if the response measurement resolves the compounds at all. We checked that first. Each exposure gives a 29-dye ΔE vector, which we centred dye-wise at each concentration and normalised to unit length, so that the comparison is between response patterns rather than response magnitudes. Agreement was then measured as the cosine between patterns, within replicates of one compound and between different compounds.

The array resolves the compounds in every run and channel (Figure 1a). Replicates of one compound correlate at $+0.345$ to $+0.542$, different compounds anti-correlate at -0.103 to -0.330, and the separation holds in all six run-channel cells, with p from 4.6×10^{-8} to

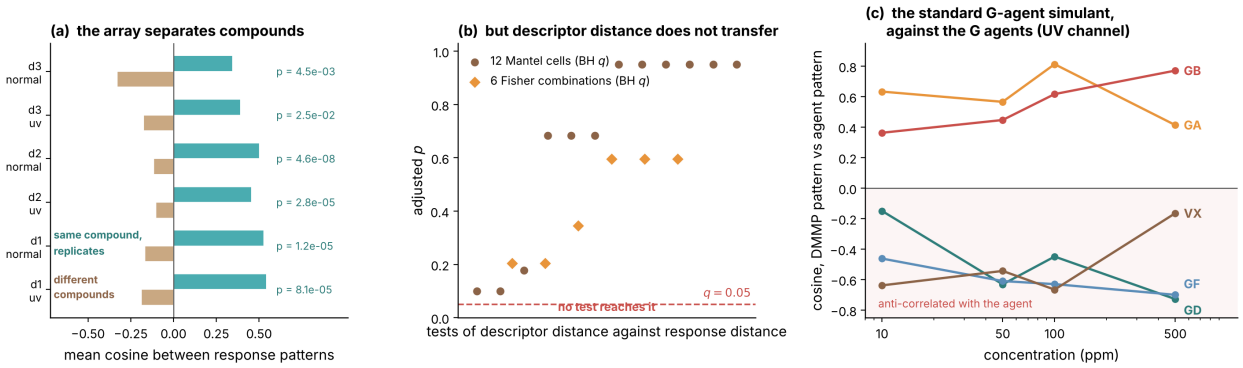


Figure 1: Response-space tests on authentic agents. (a) Mean cosine between response patterns, within replicates of the same compound and between different compounds, for three runs in two optical channels. (b) Adjusted p for every test of descriptor distance against response distance: twelve exact Mantel cells and six Fisher combinations of the independent runs. (c) Cosine between the response pattern of DMMP and those of the four G agents and VX, by concentration, in the ultraviolet channel.

0.025.

Two limits on that statement belong here rather than in the Supporting Information. The first is that dye-wise centring forces the mean between-compound cosine towards an algebraic floor of $-1/(n-1)$, and the observed means clear that floor by only 0.013 to 0.064 across the four cells where it could be evaluated. Most of the apparent separation is therefore a property of the centring, and what the test establishes is the weaker statement that replicate agreement exceeds between-compound agreement. The second is that six dye columns in the ultraviolet pivot of the second run were bit-identical across every row. Tracing them to the source images showed that those dyes do not fluoresce under ultraviolet excitation and the regions of interest had read plate background, so one channel had been recorded six times. Columns identical across all rows are counted once, which leaves 23 effective channels in that cell and 25 in the ultraviolet pivot of the third run.

Descriptor distance does not predict response distance

Simulant selection rests on an implicit transfer: that a compound close to an agent in descriptor space will be close to it in whatever the detector measures. We tested the transfer

directly, as the agreement between two distance matrices over the same compounds, one built from descriptors and one from response patterns. Agreement was measured by Mantel correlation and referred to a null built by enumerating every relabelling of the compounds, so the test is exact rather than sampled.

The test is run within each run and never across runs. The three runs share no compound, so the dye-wise sensitivity of one plate cannot be calibrated against another, and pooling responses across them would require a commutability assumption we cannot support. That restriction sets the power. Six compounds in the first run and six in the second carry both a response pattern and a descriptor vector, giving fifteen pairs in each, and the third run drops out because the A-series compounds are not in the descriptor panel. Nitrogen mustard drops out of the second run for the same reason.

Across the twelve cells, three spaces by two channels by two runs, nothing survives correction (Figure 1b). Three cells reach a nominal p below 0.05, all of them in the second run: the merged space in the ultraviolet channel at 0.0153, the structural space in the same channel at 0.0167, and the electronic space in the reflectance channel at 0.0444. Benjamini–Hochberg correction over the twelve leaves a smallest adjusted value of 0.10. The pattern of the three is itself uninformative about transfer, because the two ultraviolet cells implicate the structural and merged spaces while the reflectance cell implicates the electronic space, and the same space does not reproduce across channels of the same plate.

Combining runs does not change the answer. The first and second runs share neither compounds nor plates, so their exact Mantel p values are independent and can be combined by Fisher's method. We report all six combinations of three spaces and two channels rather than selecting one, which would reintroduce the analyst freedom the test is meant to remove. The smallest adjusted value over the six is 0.204, and none reaches 0.05.

What this does not license is the claim that no transfer exists. Fifteen pairs is a weak test, and the pooled rank correlations are correspondingly imprecise: the structural space in the ultraviolet channel gives $\rho = +0.31$ with a 95% interval from -0.45 to $+0.81$, which

excludes neither zero nor a strong association. The defensible statement is that the transfer on which simulant selection depends has not been established, in the one body of authentic-agent response data where it could be tested, and that the experiment which would settle it is a run containing a compound already measured in an existing run. Such a bridge would fix the plate correction and open the 136 cross-run pairs that are closed today.

The standard G-agent simulant against the G agents

The transfer test above treats all compounds alike. One compound deserves to be named, because the field treats it as the G-agent surrogate and because decontamination efficiency is quantified against it. The first run contains DMMP together with the four G agents and VX, so its response can be set against theirs directly.

In the ultraviolet channel DMMP correlates with tabun at $+0.764$ and with sarin at $+0.645$, and anti-correlates with soman at -0.492, cyclosarin at -0.721 and VX at -0.702. In the reflectance channel the same comparison gives -0.112, $+0.569$, -0.479, -0.495 and -0.273. The mean over the four G agents is negative in both channels, -0.189 and -0.135. The compound used in place of the G agents does not, on this measurement, resemble the G agents as a class.

A weaker claim survives in one channel. If DMMP is G-like at all it should sit closer to the G series than to VX, and that contrast is $+0.513$ with a 95% interval from $+0.131$ to $+0.722$ in the ultraviolet channel, which excludes zero. In the reflectance channel the same contrast is $+0.137$ with an interval from -0.144 to $+0.290$, which does not. One optical channel supports the ordering and the other does not resolve it.

Concentration changes the picture in the direction that matters for a detector (Figure 1c). The DMMP-to-sarin cosine rises monotonically from $+0.363$ at 10 ppm to $+0.771$ at 500 ppm, and the DMMP-to-cyclosarin cosine falls monotonically from -0.462 to -0.700 over the same range. Monotonicity is perfect in three of the ten agent-channel series. It is also the strongest result the design can give: with four concentrations the two-sided exact permutation has a

floor of $p = 0.083$, three series reach exactly that floor, and Benjamini–Hochberg correction over the ten leaves a smallest adjusted value of 0.278. We therefore report the direction and not a significance claim. The direction is that the resemblance to sarin is weakest at the low end of the range, which is the end at which point detectors are expected to alarm.

Two further readings suggested themselves to us and did not survive their own controls. We report both, because the alternative is to present artefacts as results.

The first was a ranking. DMMP is the nearest of the five candidates to tabun in the ultraviolet channel and to sarin in the reflectance channel, and it is nearest to nothing else, and the two channels never agree on which agent it is nearest to. That reads as evidence that the choice of optical channel decides which agent the standard simulant stands in for. It is not. With five candidates the probability of rank one is 1/5 under random assignment, the expected number of first places over ten agent-channel cells is 2.0, and the observed number is two, with a one-sided binomial p of 0.62. The ranking carries no information.

The second was a channel decomposition. The cosine is a sum over dye channels, and the DMMP-to-tabun value of +0.764 turns out to rest on two of the 29 channels, with the largest single channel carrying 44% of the positive mass. Read alone, that makes the resemblance look fragile. Two controls show it is not a property of DMMP. Agent-to-agent pairs in the same run concentrate identically, with median two channels in the ultraviolet and one in the reflectance channel, and a null built by permuting the dye labels of the DMMP vector gives a mean of 2.00 and 2.26. Concentration of this kind is what the product of two centred unit vectors in 29 dimensions does, and it says nothing about this compound.

What remains after both controls is the set of measured values themselves, and one structural fact. The first run contains no candidate simulant other than DMMP, so the designation cannot be contrasted against an alternative in response space at all. The permutation null that the descriptor analysis of the next section relies on is, in response space, empty.

Descriptor space cannot supply the missing validation

Neither the designation record nor a toxicity model fills the gap

Conclusions

Experimental Section

Acknowledgement

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Supporting Information Available

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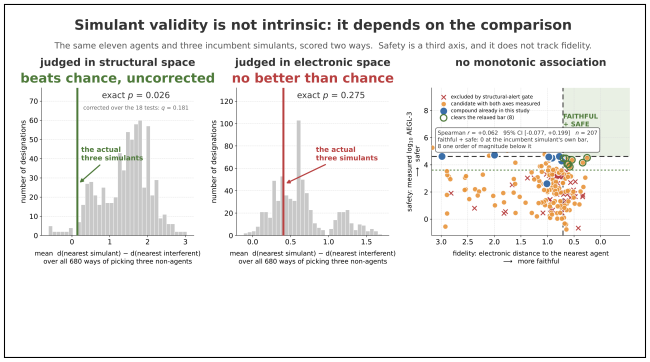
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TOC Graphic


Simulant validity is not intrinsic: it depends on the comparison
The same eleven agents and three incumbent simulants, scored two ways. Safety is a third axis, and it does not track fidelity.
judged in structural space
beats chance, uncorrected
exact p = 0.026
corrected over the 18 tests: q = 0.181
the actual three simulants
number of designations
mean d(nearest simulant) − d(nearest interferent) over all 680 ways of picking three non-agents
judged in electronic space
no better than chance
exact p = 0.275
the actual three simulants
number of designations
mean d(nearest simulant) − d(nearest interferent) over all 680 ways of picking three non-agents
no monotonic association
excluded by structural-alert gate
candidate with both axes measured
compound already in this study
clears the relaxed bar (8)
FAITHFUL + SAFE
Spearman r = +0.062 95% CI [-0.077, +0.198] n = 207
faithful + safe: 0 at the incumbent simulant's own bar;
8 nine order of magnitude below it
safety: measured log10 AEGL-3
fidelity: electronic distance to the nearest agent
→ more faithful